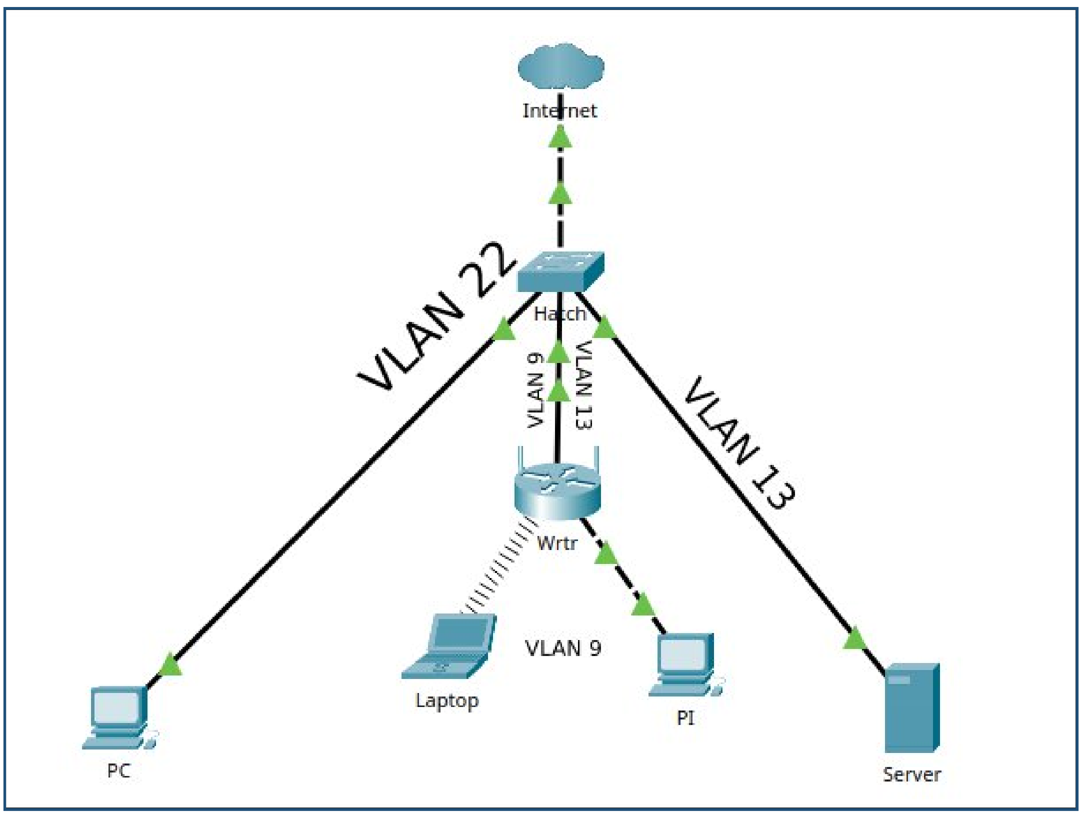
Overview

My home network is segmented in two: my Internet connection and a private lab network. The lab network is divided into three subnets and uses two VLANs. VLAN 22 is for Internet traffic and exists only on the switch for this reason. VLANs 9 and 10 use a router and trunk cable to allow for traffic between the two.

My server runs Ubuntu and has an 8TB RAID 1 setup, hosting a an NFS file share. This server uses Apache and provides DNS for the small network. SNMP is setup as well so I can see better stats in LibreNMS. LibreNMS is used for network monitoring and is running on a Raspberry Pi in a Docker container. This software monitors network statistics such as uptime, data rates, etc. The Pi also has Nessus Essentials installed for running security scans across the network.

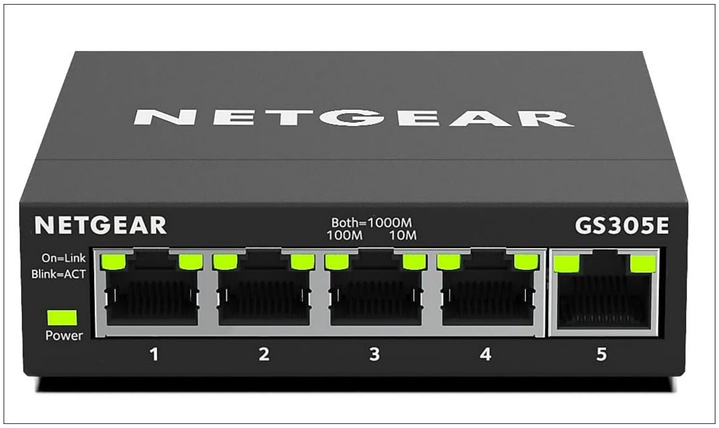
Topology



Equipment

Switch

Device: Netgear GS305E 5-Port Managed Switch



Router

Device: Archer A7 AC1750

OS: OpenWrt

VLANs

Subnets

.9

Network: 10.9.9.0/29

IP Address	Description
10.9.9.6	Default Gateway
10.9.9.5	Switch
10.9.9.4	Raspberry Pi (Monitoring)
10.9.9.3	Laptop
10.9.9.2	VM (Main PC)
10.9.9.1	Main PC

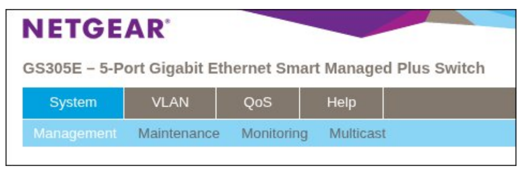
.13

Network: 10.9.9.8/29

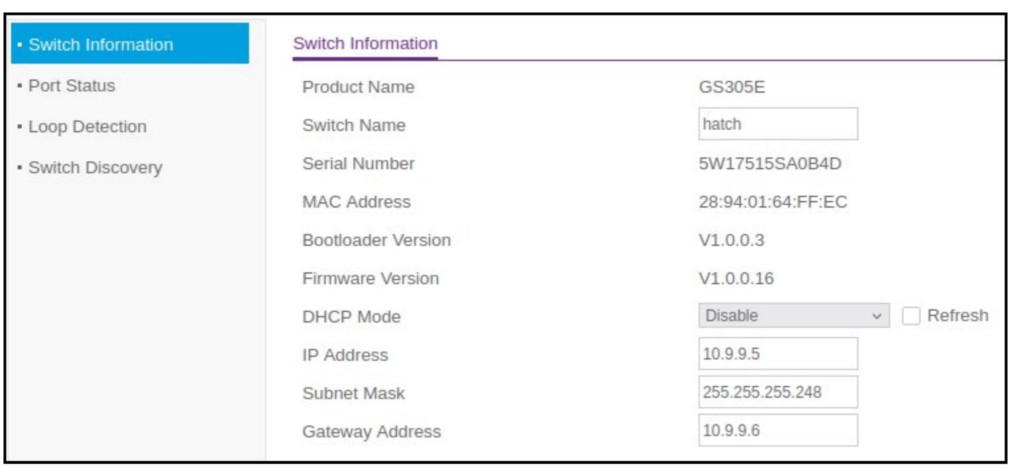
IP Address	Description
10.9.9.9	Default Gateway
10.9.9.10	Server

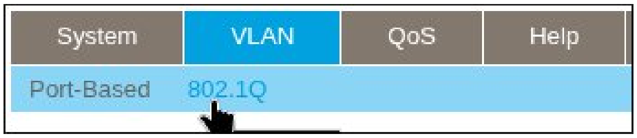
Configuring VLANs

Switch

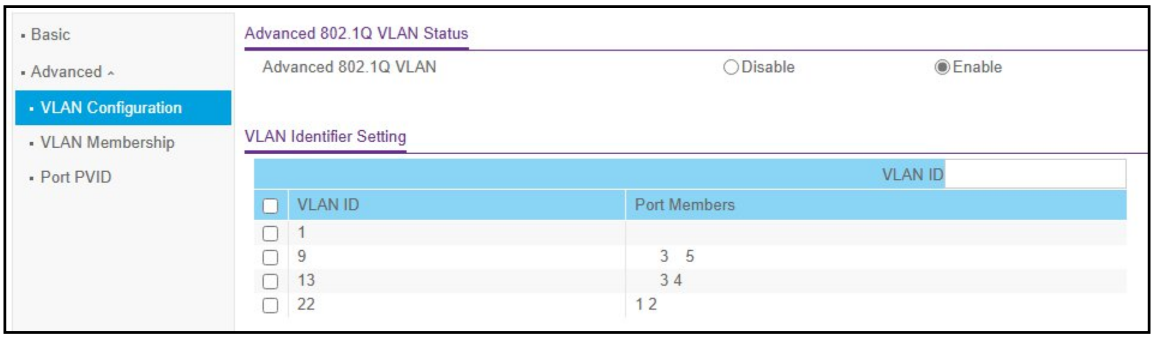


Management for the Netgear GS305E is done through a GUI via web browser. In the top-left of the page is the toolbar. Each tab has its own options. I only need to set an IP address and configure VLANs. The IP address can be configured on the first page after logging in. The below screenshot shows the landing page. For my setup I disabled DHCP since it will be handled by the router. I set the IP address, subnet mask, and gateway address as shown in the screenshot.

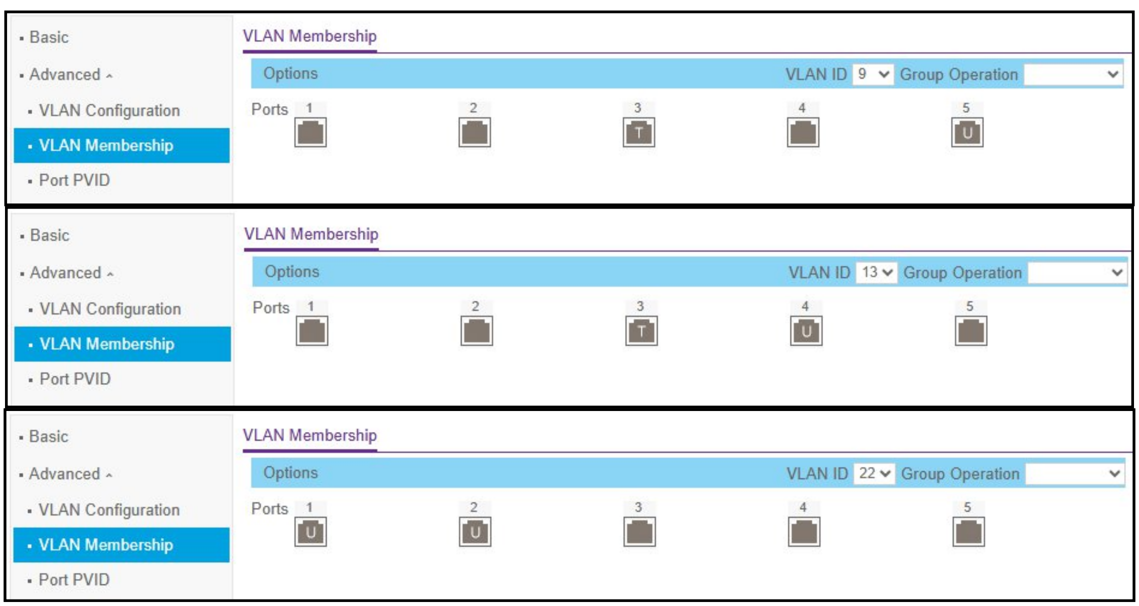




To get to VLAN management I went to the VLAN tab and then 802.1Q. In the left pane I selected Advanced and set the option to Enable. Doing this opens up all of the advanced options as shown below. The VLAN Configuration page gives a good overview of all of the VLANs and their members. Typing a number in the VLAN ID box and clicking the apply button will create a new VLAN with that number.



The VLAN Membership page allows you to adjust membership and choose whether port traffic is tagged or untagged. Untagged traffic can transmit freely within its associated VLAN. In my setup the Tagged port (Port 3) will be used for a trunk cable. This port is set as Tagged in both VLAN 9 and VLAN 13. Later the router will be setup to route traffic between these two VLANs using the trunk.

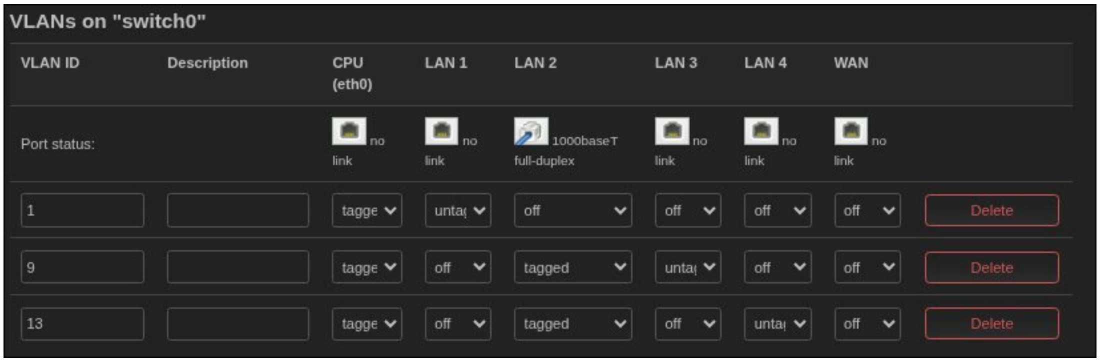


*** NOTE ***

Not shown in the screenshots: VLAN 1 is reserved for a single port on both the switch and Router. These ports are meant to act as emergency management ports in case of a misconfiguration during testing.

Router

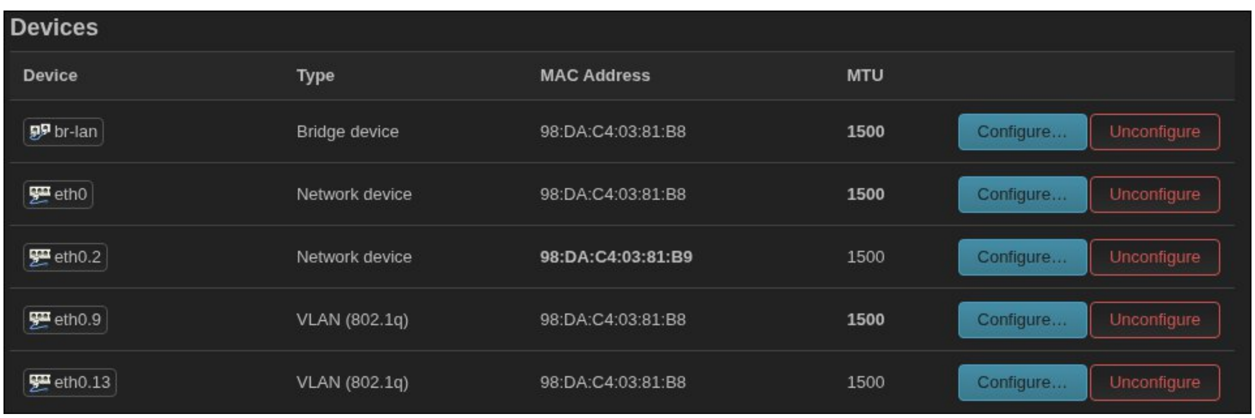
The router in my setup is running OpenWrt, a Linux based operating system for routers. It allows for much more control over the router than the standard firmware. First I configured the router's switch using the web interface, which is very similar to the Netgear's.


VLANs on "switch0"

VLAN ID	Description	CPU (eth0)	LAN 1	LAN 2	LAN 3	LAN 4	WAN	
Port status:		no link	no link	1000baseT full-duplex	no link	no link	no link	
1		tagge	unta	off	off	off	off	Delete
9		tagge	off	tagged	unta	off	off	Delete
13		tagge	off	tagged	off	unta	off	Delete

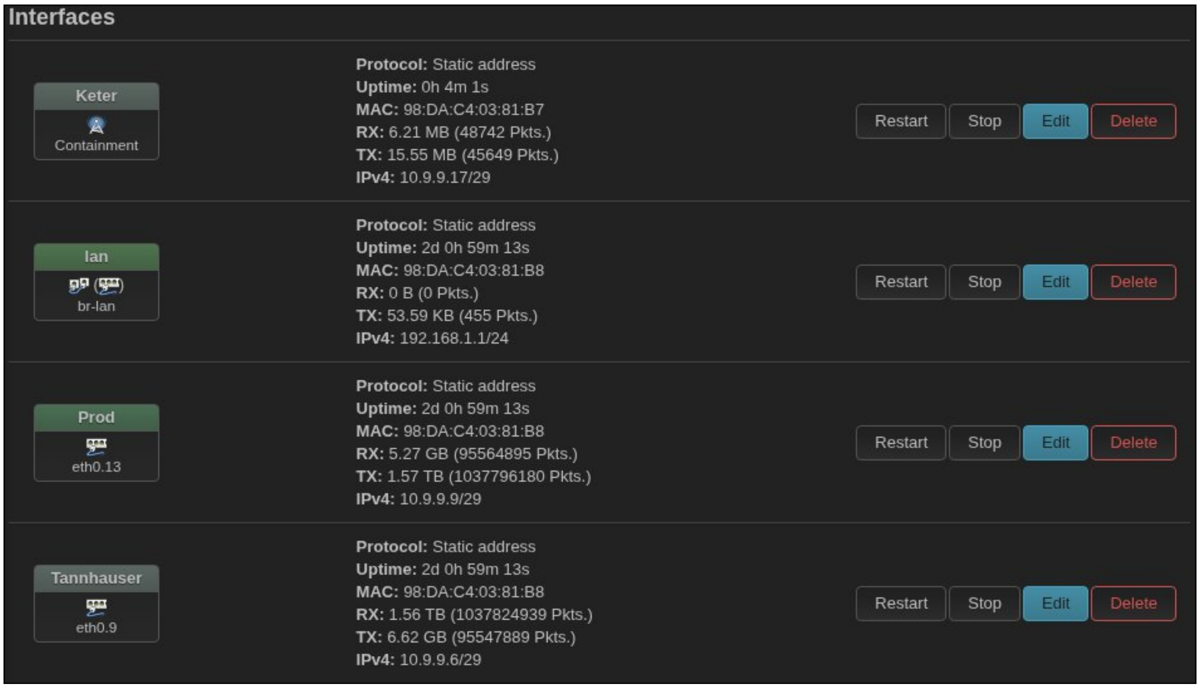
The main thing to note about the router's switch is that the CPU can be marked as tagged or untagged. If it's set to *untagged* it allows for routing within that VLAN. *Tagged* CPU traffic allows for routing between VLANs. A trunk interface/cable is still necessary to successfully route traffic between VLANs.

Once a VLAN has been created on the router's switch a device is created it (**eth0.9** & **eth0.13**). Devices control physical aspects of the interfaces. You can define the maximum transmission unit (MTU), create the 802.1Q ports, and more. To finish setting up the VLANs on the router an interface is required for each. These interfaces function slightly different than the tradition interface on something like a Cisco router.


Devices

Device	Type	MAC Address	MTU		
br-lan	Bridge device	98:DA:C4:03:81:B8	**1500**	Configure…	Unconfigure
eth0	Network device	98:DA:C4:03:81:B8	**1500**	Configure…	Unconfigure
eth0.2	Network device	**98:DA:C4:03:81:B9**	1500	Configure…	Unconfigure
eth0.9	VLAN (802.1q)	98:DA:C4:03:81:B8	**1500**	Configure…	Unconfigure
eth0.13	VLAN (802.1q)	98:DA:C4:03:81:B8	1500	Configure…	Unconfigure

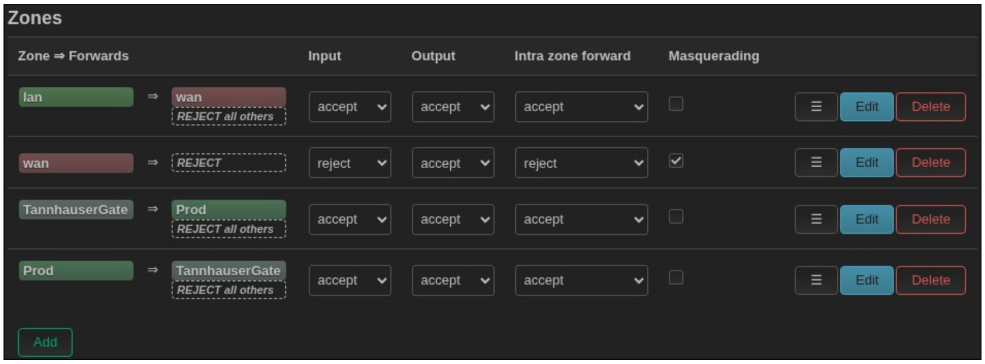
Below is a screen shot of the OpenWrt *interfaces*. I named these interfaces **Tannhauser** and **Prod**. These interfaces are where network information is configured. You can manually set IPv4 and IPv6 addresses, setup a DHCP server for that interface, and choose the device that interface maps to. A single port, group of ports, or 802.1Q configured port can be mapped to an OpenWrt interface.



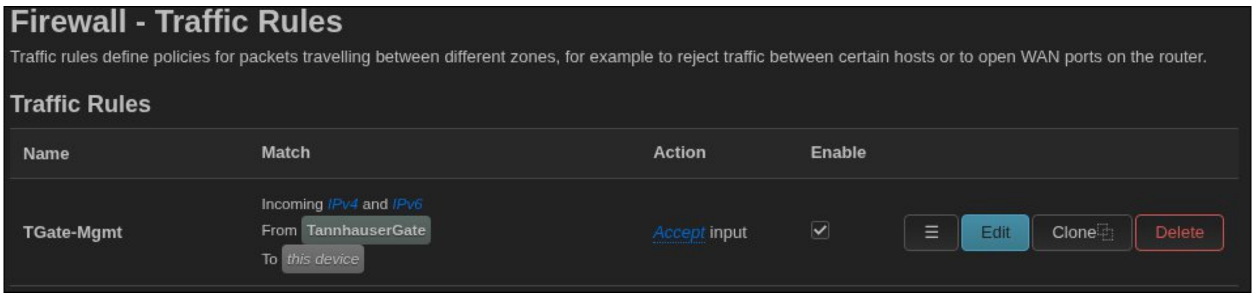
I chose the private network address 10.9.9.0/24 and decided to subnet that as needed. VLAN 9 uses the address **10.9.9.0/29** and VLAN 13 uses the address **10.9.9.8/29**. Any other subnets I use will build off of this addressing scheme (i.e. 10.9.9.16/29). The idea is to keep the number of hosts as low as possible while leaving room for experimenting with different devices.

Firewall

Lastly, the firewall needs to be setup for traffic to route properly between VLANs. This is done by adding the different interfaces to a firewall zone. These zones are then setup to forward traffic as shown in the following screen shot.



The *type* of traffic can then be limited using firewall rules. I added a rule to allow devices on VLAN 9 to manage the router.

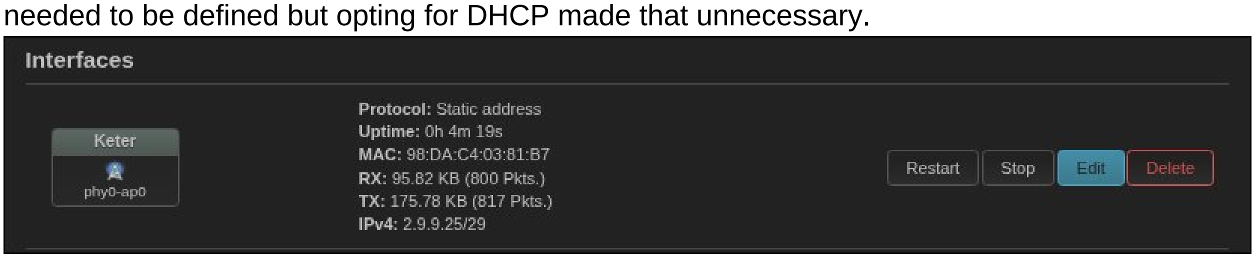


Wireless

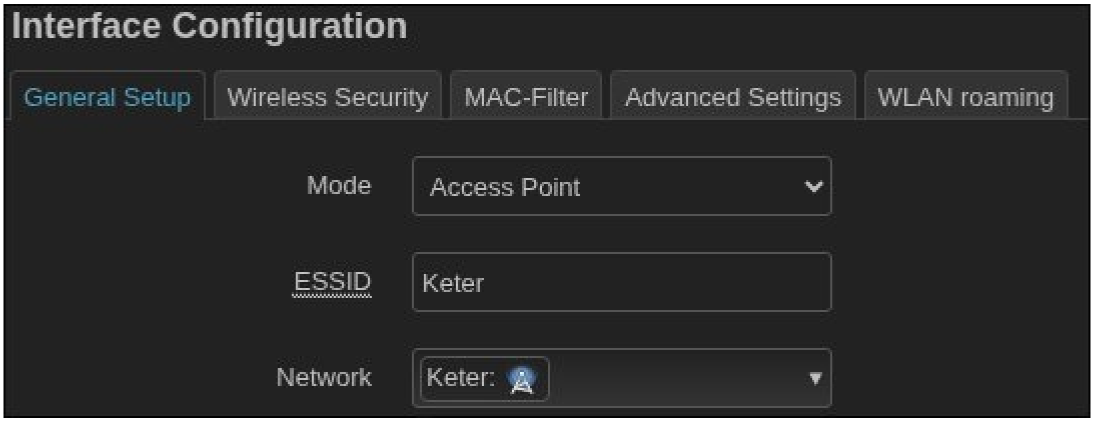
Setting up a wireless interface is almost identical to setting up the physical ones. Under the wireless settings of the router are the two different wireless options (2.4Ghz/5Ghz). Multiple wireless networks can be created for each but only one can be used as the current profile for that transmitter. The wireless network address is **10.9.9.16/29** with a hidden SSID. I used the 5Ghz transmitter and also lowered the power level to decrease it's visibility outside of the immediate area.

Create Device & Interface

An interface was created for the wireless network. I prefer using static addressing in my home network since it's small and manageable. I can easily identify my devices based on IP. For the wireless traffic to route properly, however, I needed to enable DHCP. A static route needed to be defined but opting for DHCP made that unnecessary.



After the *interface* is configured a device is created for the wireless network. Here you choose a *network* to map the device to. The *network* refers to the *interface*, which has the network information. I setup WPA3 authentication and enabled the setting to hide the ESSID to hide the network from my neighbors. I also lowered the transmit power to reduce the range.



The last thing to do for the wireless interface is to make sure it's a member of the Tannhauser firewall zone. With everything configured, the wireless network can now contact the server.